

Distilling a Resource Constrained Rotary Language Model

DASE7506 Mini Project 1 Small Language Model Challenge

Final frozen evaluation September 2026

Final test BPB 1.519497 | 6.59M parameters | CPU ratio 4.1547x

Abstract

This project improves the supplied randomly initialized WikiText-2 language model while preserving the fixed BPE-2048 tokenizer, causal windowing, and CPU FP32 evaluator. I first isolate rotary position embeddings (RoPE) and SwiGLU in a paired 2x2, three-seed ablation. I then search capacity under the 5x CPU-time, 4 GiB RAM, and 64 MiB asset limits. The final predictor is a width-288, six-layer RoPE+SwiGLU student distilled from two independently initialized width-448 teachers and averaged over four predeclared late checkpoints. It achieves 1.494533 validation BPB and 1.519497 test BPB, compared with 2.101258 test BPB for the original baseline. The student-only bundle remains within every deployment constraint.

1 Task and experimental protocol

The task is next-token prediction with context 256. All trainable models start from random initialization and learn only from the supplied training split. Validation is used for predeclared checkpoint and configuration selection; it never supplies gradients. The test split was not read until the predictor, averaging rule, configuration, and assets were frozen.

Item	Fixed protocol
Training	GPU FP32; effective batch 32; AdamW; 100-step warm-up; cosine decay; weight decay 0.1; gradient clipping 1
Evaluation	Official scorer; CPU FP32; four threads; complete independent causal windows
Randomness	Independent model, data-sampler, and dropout RNG streams; recorded resume states
Primary metric	Bits per UTF-8 byte (BPB); lower is better

2 Architecture and controlled ablation

2.1 RoPE and SwiGLU

RoPE rotates the full query/key head dimension with base 10,000 and removes the learned position table. SwiGLU replaces the GELU feed-forward block. Its hidden width is approximately $8d/3$, rounded to a multiple of eight, keeping the dominant feed-forward matrix count close to the GELU baseline. Token embedding and output weights remain tied.

2.2 Three seed 2x2 experiment

A, B, C, and D use width 128, depth 4, context 256, and exactly 1,200 updates (9,830,400 processed targets) for seeds 17, 42, and 123. Within each seed, all variants receive the identical sampled-window sequence.

Group	Position	FFN	Mean validation BPB	Difference from A
A	Learned	GELU	2.0749	0.0000
B	RoPE	GELU	1.9217	-0.1533
C	Learned	SwiGLU	2.0083	-0.0666
D	RoPE	SwiGLU	1.8336	-0.2413

The descriptive interaction $D - C - B + A$ is -0.0215 BPB. Negative values favor the modified model, so both mechanisms help and their combination slightly exceeds the sum of their isolated gains. With only three seeds, these results are not presented as a significance test.

Architecture ablation across three seeds

Validation BPB after 1,200 updates; points are seeds 17, 42, and 123; diamonds show mean $\pm$ sample SD

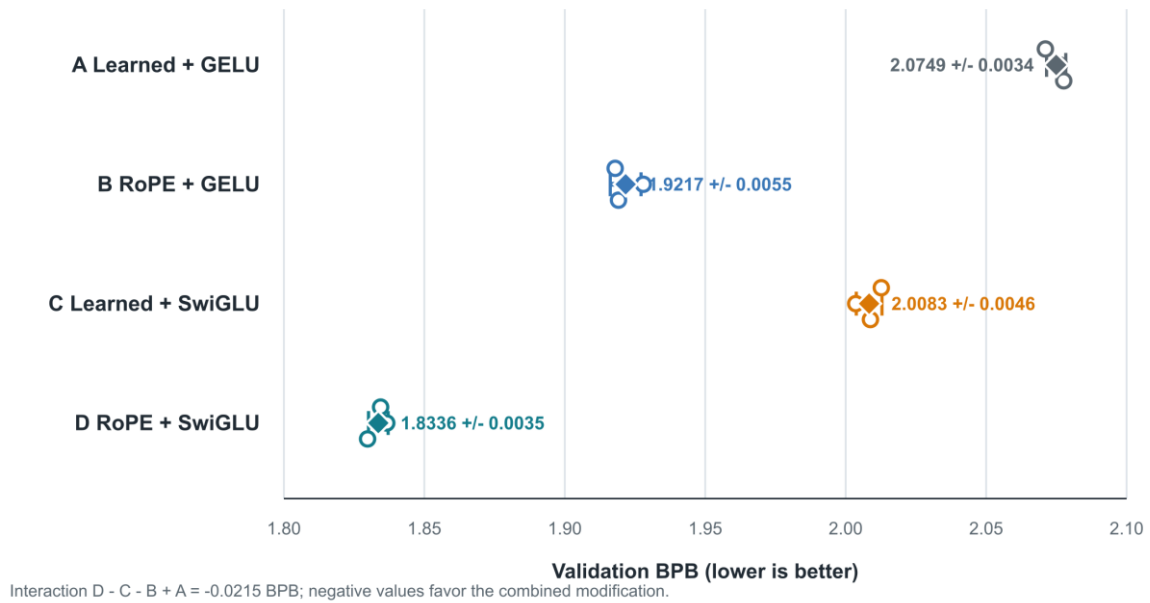


Figure 1. Three-seed architectural ablation at 1,200 updates. Points are individual seeds; bars show mean $\pm$ sample standard deviation.

Figure 1 makes the paired pattern visible beyond the means in the table. RoPE provides the larger isolated reduction (0.1533 BPB versus 0.0666 for SwiGLU), while D improves on A for every seed and has a mean gain of 0.2413 BPB. The seed clusters are narrow relative to these effects, but the sample remains too small for a claim of statistical significance. This evidence motivates carrying the combined RoPE+SwiGLU architecture into the capacity search.

3 Capacity search and teacher construction

Seed-17, 1,200-update screens covered widths 192, 224, 256, 288, 320, and 384 at depth six. Validation BPB improved as capacity increased, while CPU cost approached the 5x limit. Clean-system resource timing qualified width 288 near the internal 4.5x safety line. Width 448 was therefore used only as a training-time teacher and is absent from inference assets.

Capacity improves BPB but raises CPU cost

D models, seed 17, depth 6, 1,200 updates; labels show scale and initial validation CPU ratio

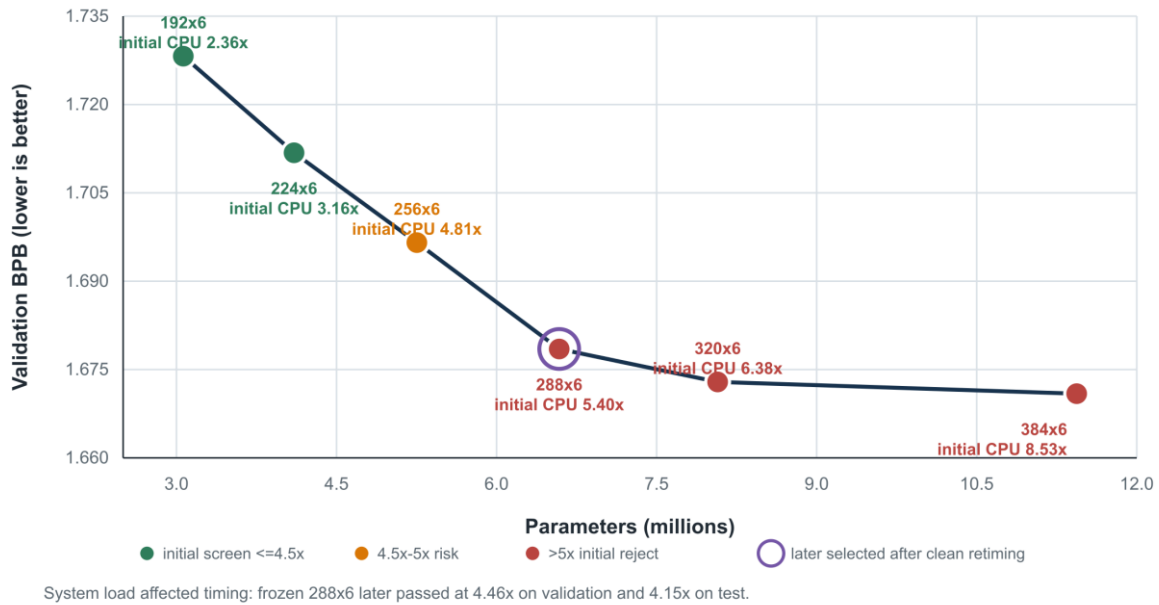


Figure 2. Capacity-quality-resource trade-off for six-layer RoPE+SwiGLU screens. Initial CPU ratios are shown alongside the clean retiming decision.

Scaling improves validation BPB throughout the measured grid, but the quality gain becomes increasingly expensive near the deployment boundary: moving from width 288 to 384 improves the screen by only about 0.0076 BPB while the initial CPU ratio rises from 5.40x to 8.53x. Host load materially affected the early timing screen, so feasibility was not inferred from that isolated run. A matched clean retiming qualified width 288 at 4.46x on validation and the frozen test measurement later confirmed 4.15x. Width 288 was therefore selected as the largest practical student, while larger models remained eligible only as teachers.

Model	Purpose	Training	Selected validation BPB
D-448x6, seed 17	Teacher 1	6,000 updates; selected step 2,400	1.618143
D-448x6, seed 42	Teacher 2	6,000 updates; selected step 2,400	1.624628
D-288x6	Deployable student	6,000 updates	1.500414 (single checkpoint)

Both teachers were independently initialized and trained only on supplied training text. They generalized best at step 2,400 and overfit thereafter. Teacher selection used validation only; teacher checkpoints are not required by the final predictor.

4 Distillation and final matched experiment

4.1 Objective and teacher mixture

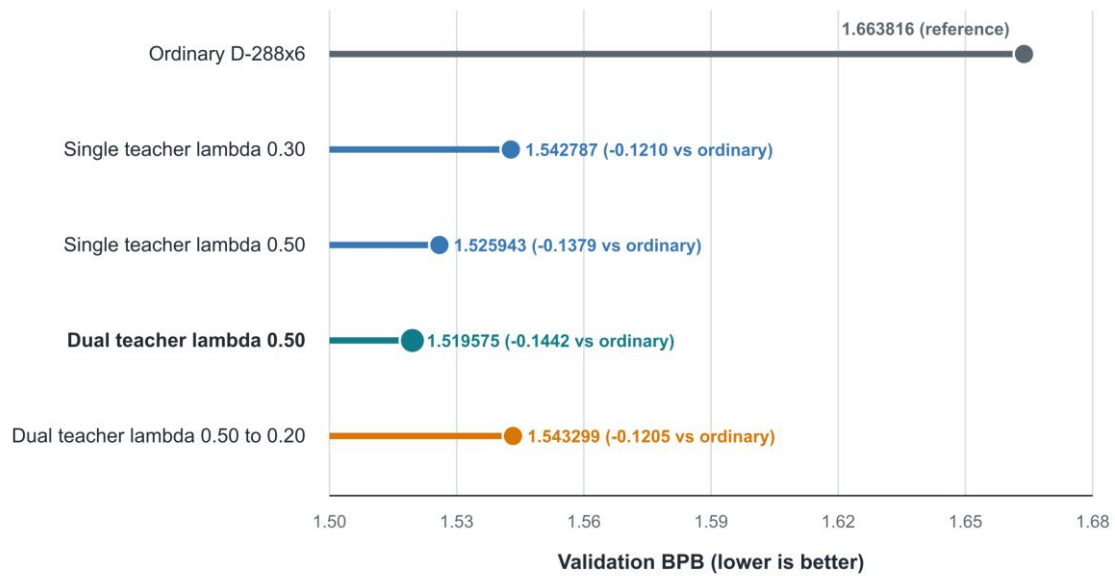
For temperature $T=2$, the student minimizes $(1-\lambda)$ times the hard-label cross-entropy plus λ times T squared times KL divergence from the teacher distribution. For two teachers, temperature-scaled probabilities are averaged 50:50 before the KL term. Teachers are frozen and receive no gradients.

Student pilot	Updates	Validation BPB	Decision
Single teacher, lambda 0.30	3,600	1.542787	Rejected
Single teacher, lambda 0.50	3,600	1.525943	Best single teacher
Two teachers, lambda 0.50	3,600	1.519575	Selected
Two teachers, lambda 0.50 to 0.20	3,600	1.543299	Annealing rejected

The dual-teacher pilot improves on the best single teacher by 0.00637 BPB, exceeding the predeclared 0.002 expansion threshold. A later linear-lambda pilot was worse and was retained as a negative result.

Distillation pilot comparison

D-288x6, seed 17, 3,600 updates; every method uses the same inference architecture



Dual teachers improved over the best single teacher by 0.00637 BPB; annealing the distillation weight was worse.

Figure 3. Distillation pilot comparison for the same D-288x6 student architecture after 3,600 updates.

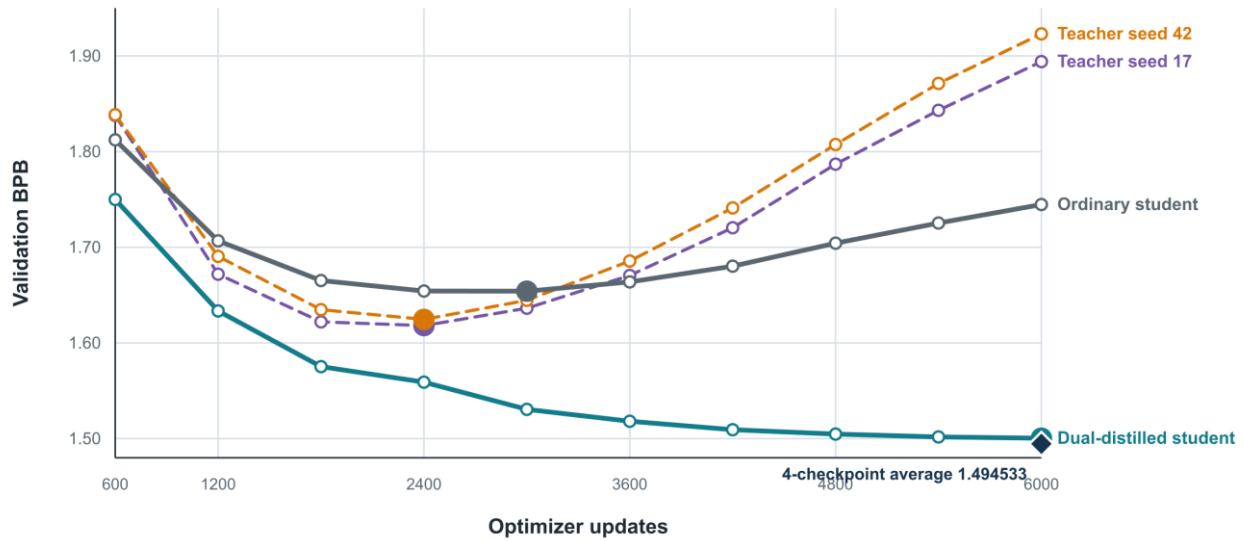
Because every bar uses the same student inference architecture and training budget, the differences isolate the training target rather than deployment capacity. Increasing lambda from 0.30 to 0.50 improves the single-teacher result by 0.01684 BPB, and adding a second independent teacher yields a further 0.00637 BPB. The annealed schedule loses 0.02372 BPB relative to constant dual-teacher training. These results select the simpler constant $T=2$, $\lambda=0.50$, 50:50 dual-teacher objective for the final run.

4.2 Matched 6000 update comparison

The ordinary and distilled D-288x6 students were initialized from scratch with seed 17 and trained on the same sampled-window schedule. The ordinary model selected step 3,000 at 1.654126 validation BPB and degraded thereafter. The distilled model continued improving to 1.500414 at step 6,000. At the strictly matched step-3,000 endpoint, distillation improves BPB by 0.1235 without changing inference capacity.

Validation dynamics reveal overfitting and distillation gains

Teachers and ordinary student overfit; the dual-distilled student continues improving through 6,000 updates



Filled markers indicate the validation-selected checkpoint for each run; the average was fixed before test evaluation.

Figure 4. Validation trajectories of the two teachers, the ordinary student, and the dual-distilled student. Markers identify selected checkpoints.

The trajectories explain why a fixed final-step comparison alone would be misleading. Both teachers reach their minima at step 2,400 and then degrade; the ordinary student similarly bottoms out at step 3,000. In contrast, the distilled student continues improving through step 6,000. This pattern is consistent with the soft targets providing useful regularization, although the final long-run comparison uses one student seed and cannot establish that mechanism statistically.

4.3 Checkpoint averaging

The predeclared arithmetic mean of student checkpoints at steps 4,200, 4,800, 5,400, and 6,000 was accumulated in FP64 and exported in FP32. It reaches 1.494533 validation BPB, improving on the best single checkpoint by 0.00588. Two cheaper subsets scored 1.497331 and 1.495849 and therefore did not replace the four-point average.

5 Final score resources and reproducibility

Quantity	Baseline	Frozen student	Limit or outcome
Test BPB	2.101258	1.519497	Lower is better
CPU test scorer time	12.5708 s	52.2281 s	4.1547x; pass (<5x)
Peak test process-tree RAM	-	1.74 GiB	Pass (<4 GiB)
Uncompressed inference assets	-	27.54 MiB	Pass (<64 MiB)
Parameters	-	6,587,136	Student only

The frozen checkpoint SHA-256 is 5414d777646765f4429b7f59af832edb92df2acb5e17d32cd33a41d67d191b4e. Its bundle contains only the student and inference dependencies. Teachers, optimizer states, and resume states are excluded. Repeated candidate test scores were bit-for-bit identical; repeats supported resource diagnostics and a final user-requested reproducibility check, and never influenced selection.

6 Cost failure accounting and limitations

The campaign contains 38 formal run directories and 105,600 optimizer updates. Recorded model training and full-validation alone required 7.065 hours: 5.241 GPU training hours and 1.824 CPU validation hours. This excludes environment setup, diagnostics, unit tests, checkpoint averaging, resource qualification, test reproduction, and reporting. All failed launches, rejected models, teacher overfitting, an initially incomplete bundle dependency, and invalid RAM-monitor attempts are retained in the cost ledger rather than silently discarded.

The main limitation is that capacity and distillation selection rely mostly on seed 17; the final gain is not a multi-seed uncertainty estimate. The capacity grid is bounded and does not prove global optimality. The teachers overfit after step 2,400, and the distilled student inherits any biases in their probability targets. CPU timing also depends on thermal state and host load, so compliance is based on matched runs and a safety margin, not an isolated fast measurement. The first RAM monitors observed only the virtual-environment launcher; calibrated parent-and-child monitoring corrected the resource record.

7 Attribution and AI assistance

RoPE follows Su et al. (2021); SwiGLU follows Shazeer (2020); knowledge distillation follows Hinton, Vinyals, and Dean (2015). Checkpoint averaging is a local fixed elementwise arithmetic mean. No external training text, pretrained weight, cached test answer, or evaluation-time network access is used.

AI assistance was used to discuss the feasibility of candidate methods, correct conceptual misunderstandings, and support implementation and experimental verification. All model weights were trained locally from random initialization using only the supplied training split.

References

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